

Master Degree in Artificial Intelligence for Science and Technology

Unsupervised Learning



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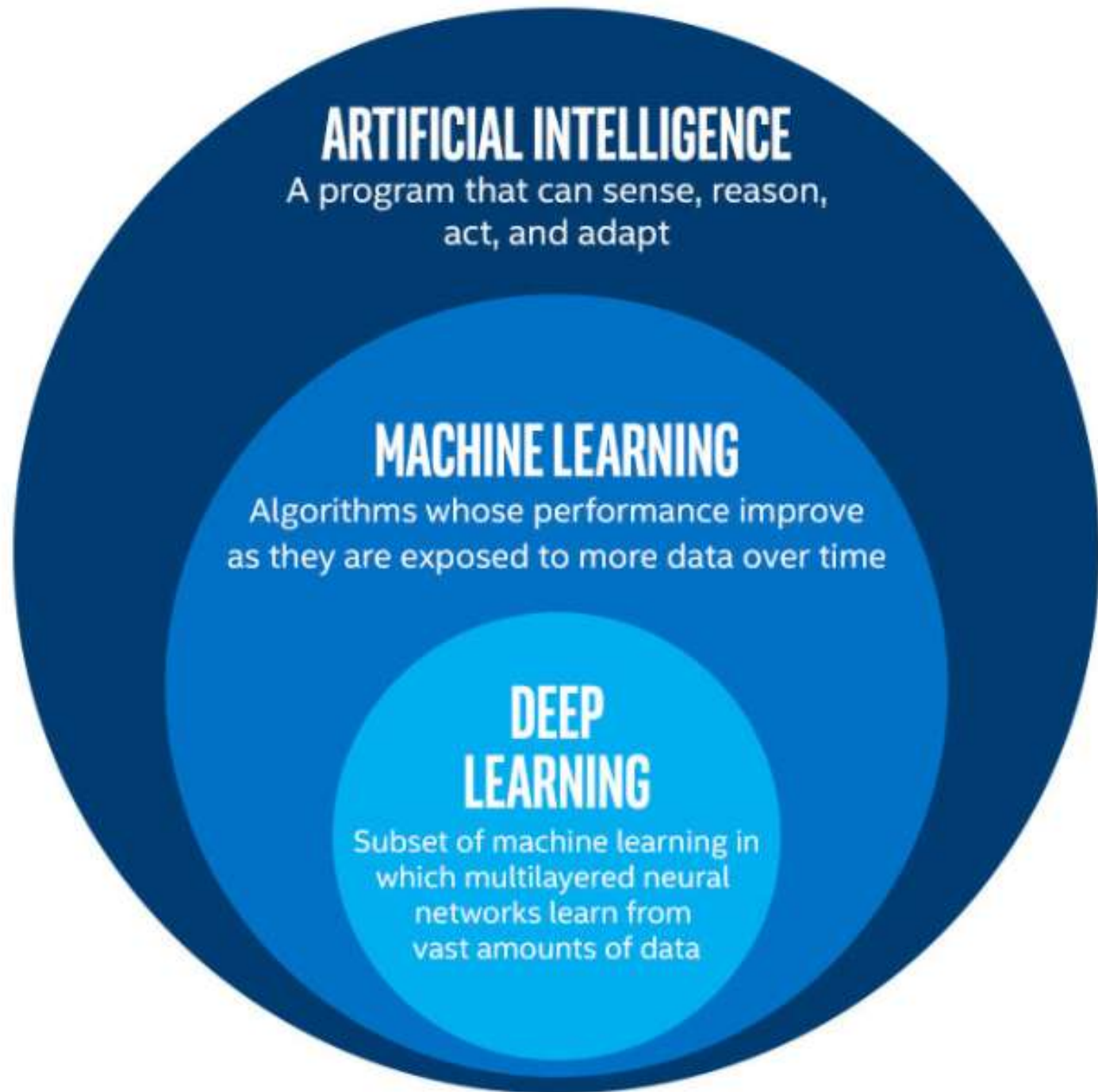
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Artificial Intelligence;

intelligence demonstrated by machines, in contrast to the natural intelligence displayed by humans.

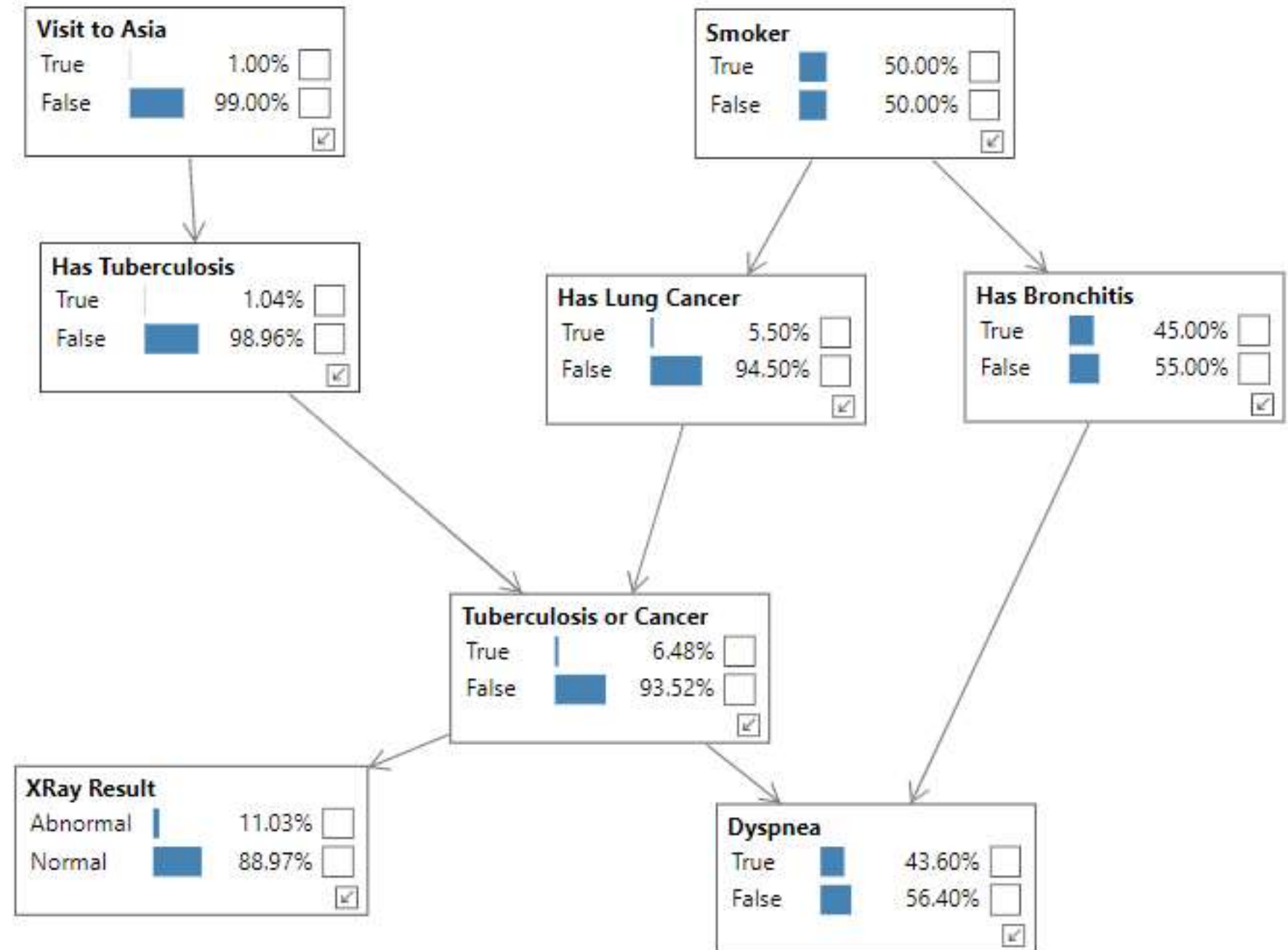
- Colloquially, the term Artificial Intelligence (**AI**) is used to describe machines/computers that mimic “cognitive” functions that humans associate with other human minds, such as “learning” and “problem solving”.
- Two kinds of AI:
 - ✓ Weak
 - ✓ Strong



Artificial Intelligence

Bayesian Networks

- A type of statistical model that represents a set of variables and their conditional dependencies via a directed acyclic graph (DAG).
- Bayesian networks are ideal for taking an event that occurred and predicting the likelihood that any one of several possible known causes was the contributing factor.
- Structural Causal Models.

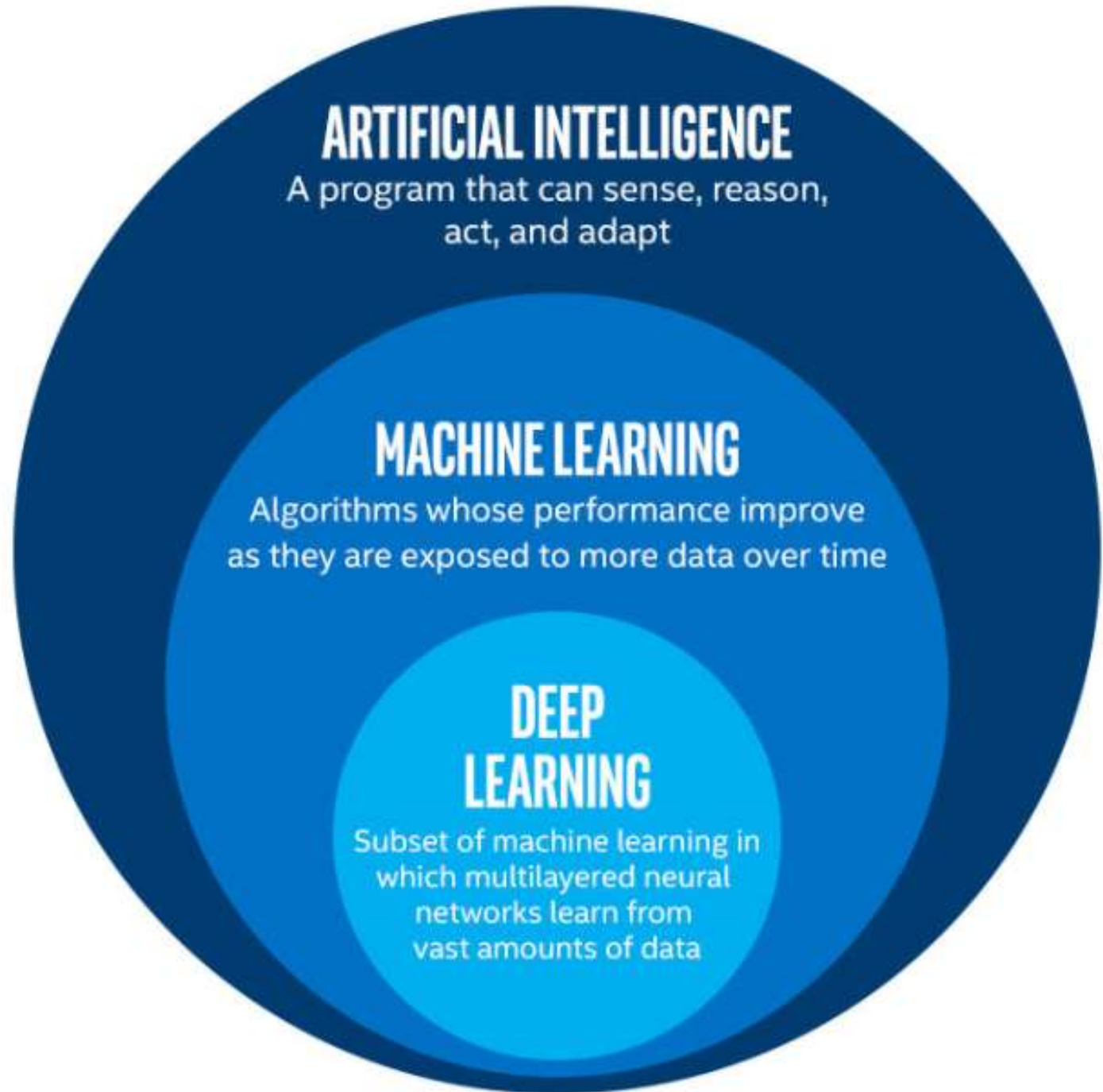


Machine Learning

algorithms and statistical models that computer systems use in order to perform a specific task effectively without using explicit instructions, relying on patterns and inference instead.

Three kinds of ML;

- Supervised
- Unsupervised
- Reinforcement Learning

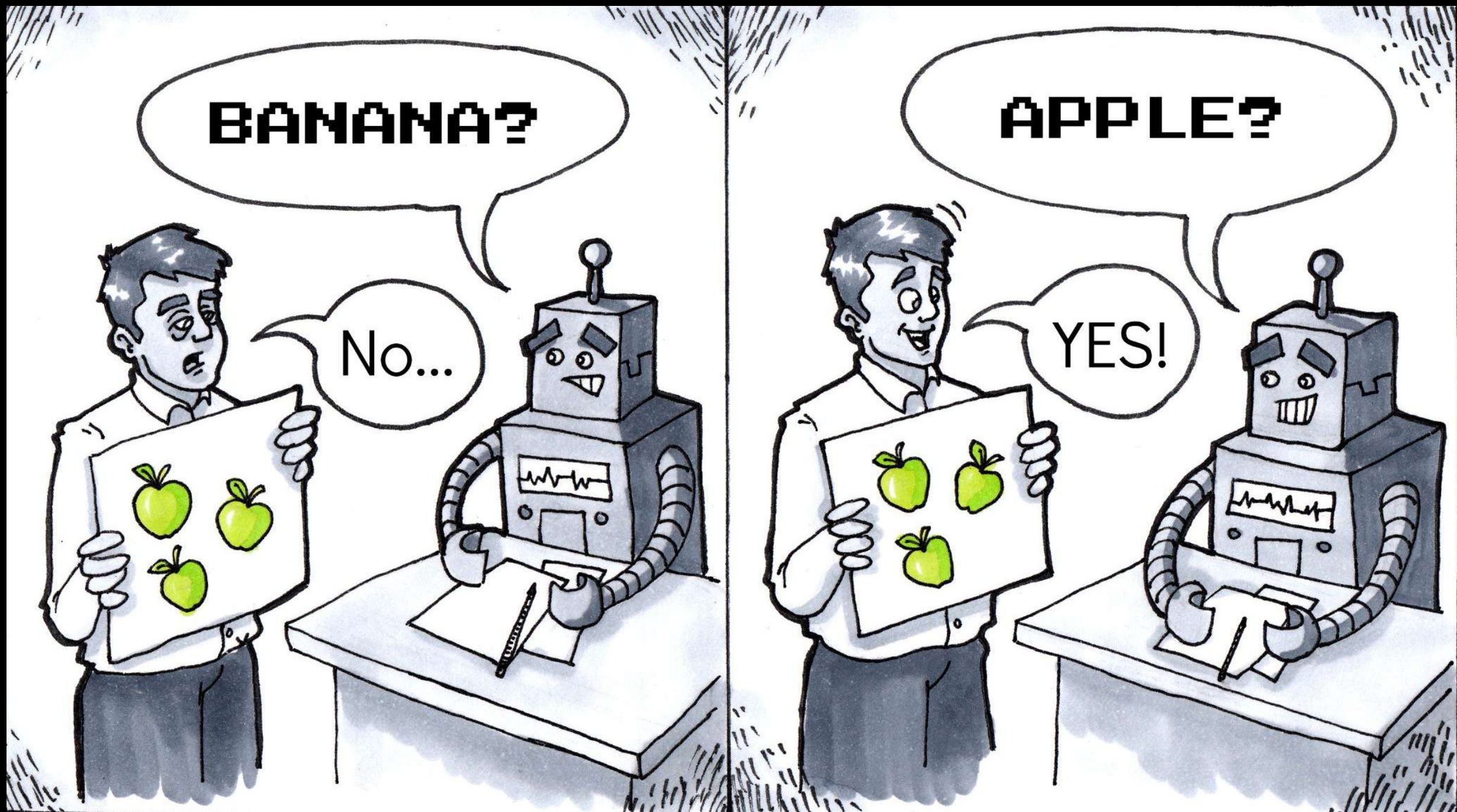


BANANA?

No...

APPLE?

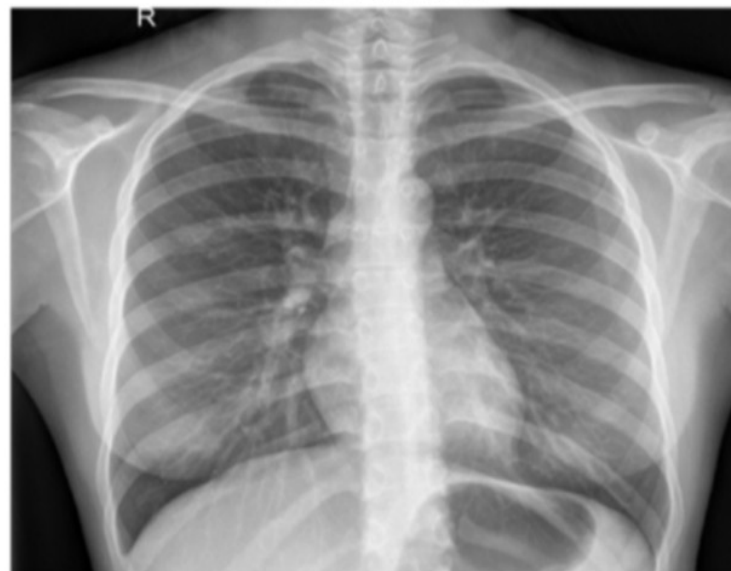
YES!



Normal



Normal



Normal



COVID-19

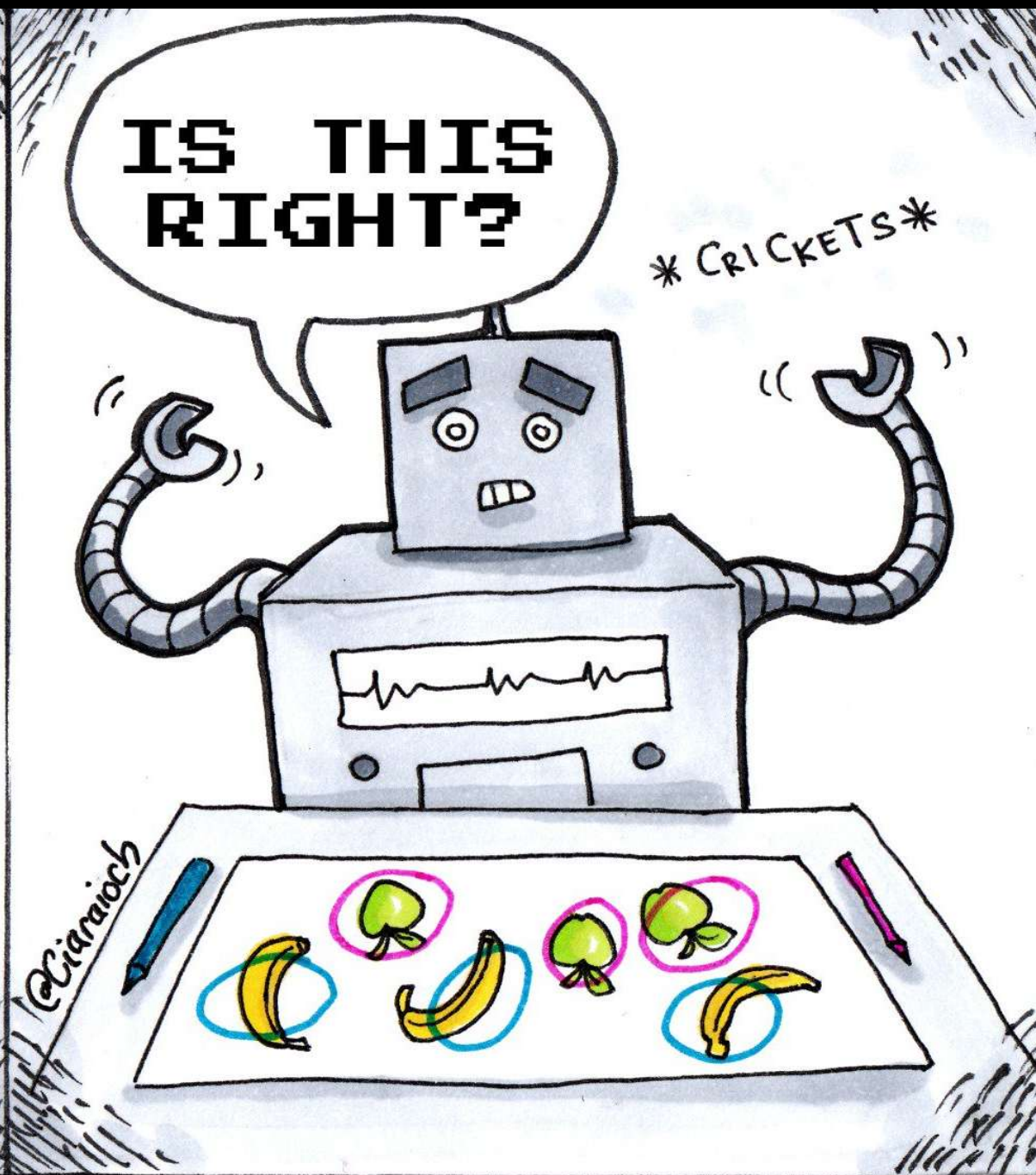
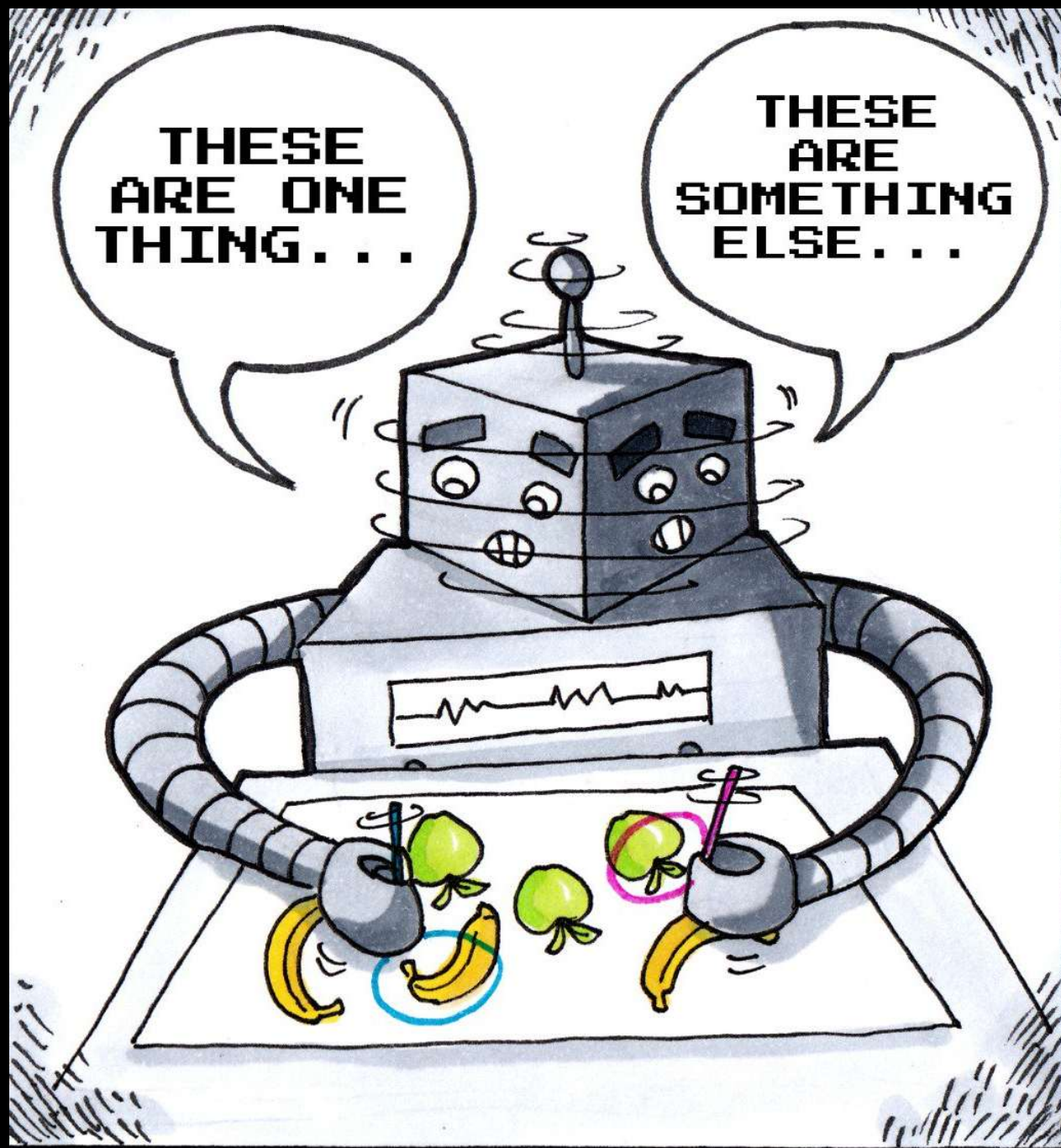


COVID-19



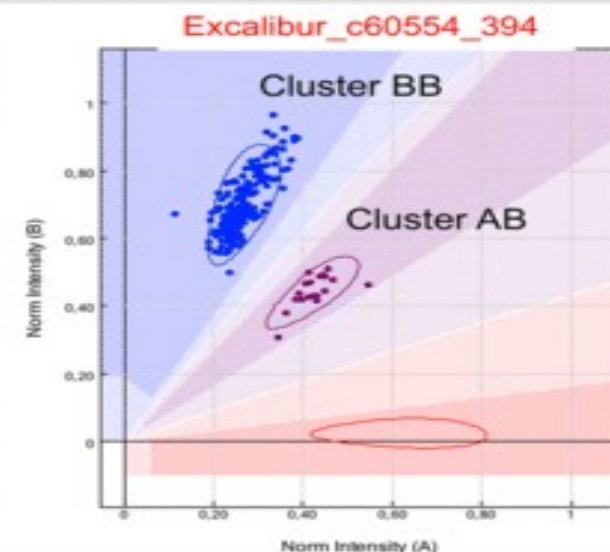
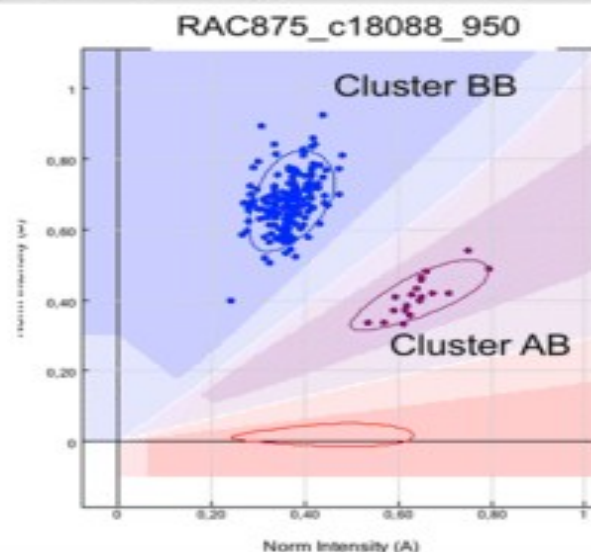
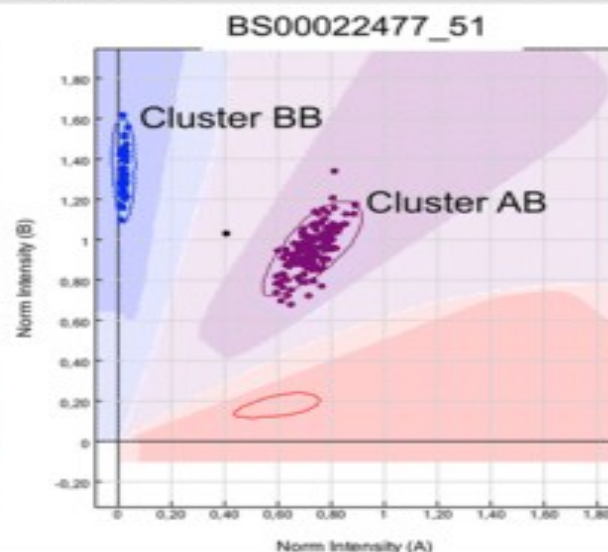
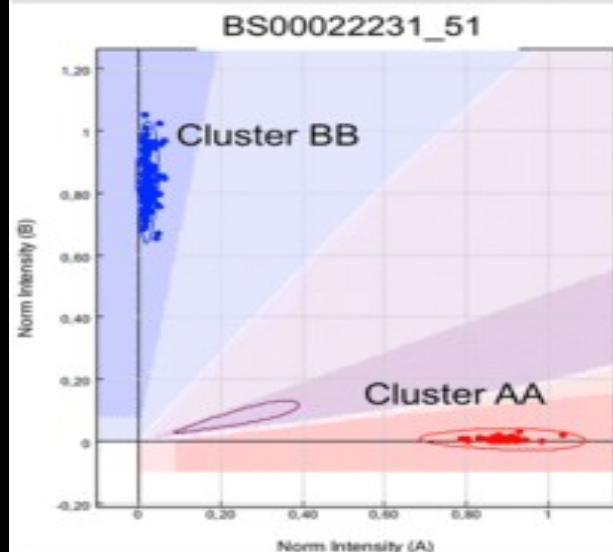
COVID-19





(a) Simple SNP**(b) Biallelic hemi SNP****(c) Biallelic hemi SNP****(d) Triallelic hemi SNP**

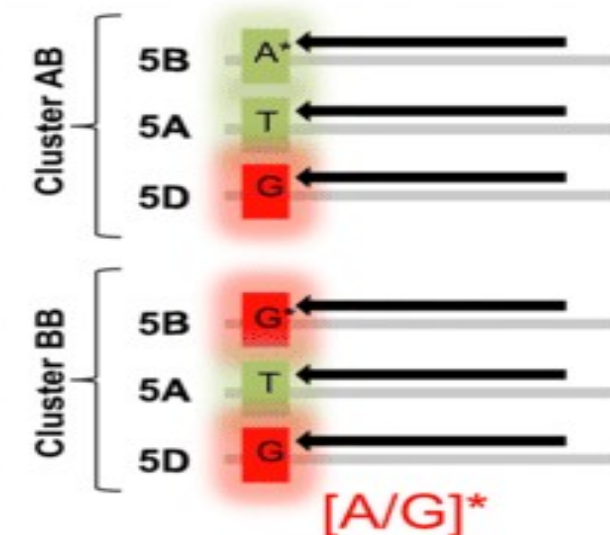
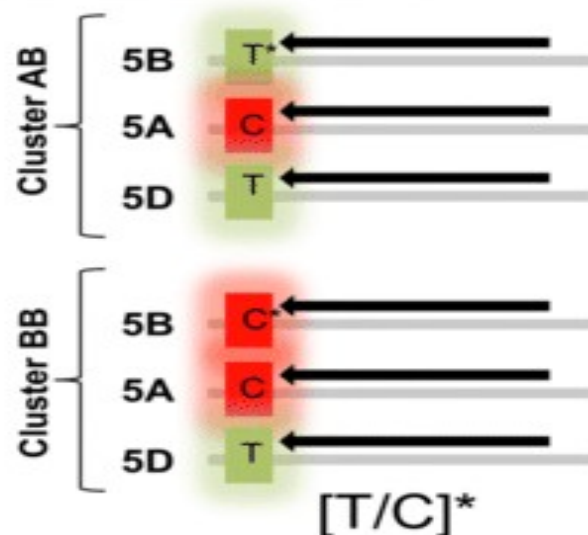
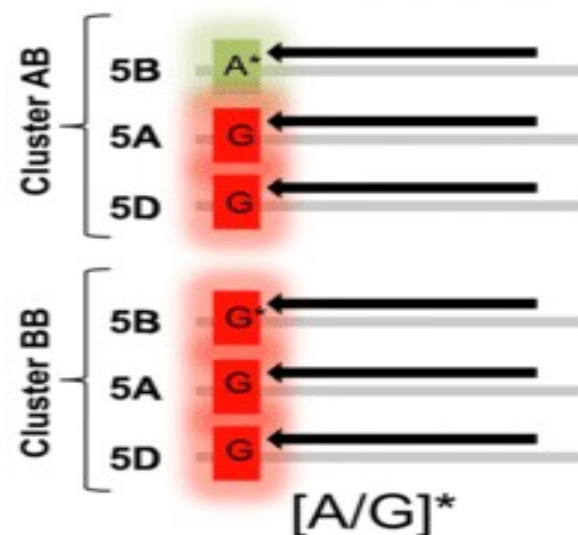
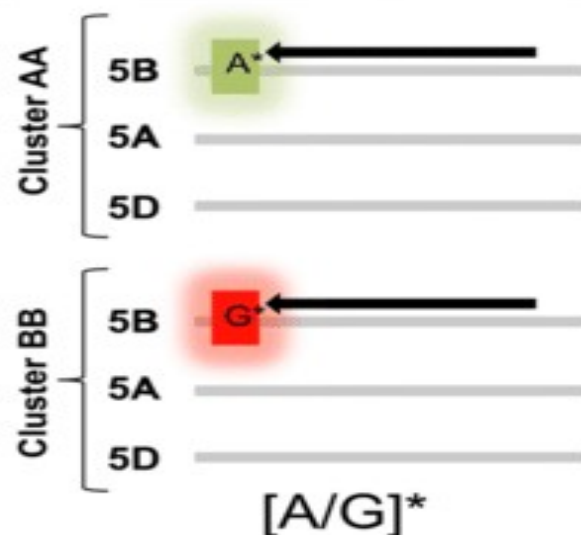
Clustering of 90k Infinium SNP array raw data by Illumina GenomeStudio



5B homeolog-specific SNP calls predicted by TraitGenetics using their own cluster file

[A/G]**[A/G]****[T/C]****[T/G]**

SNP identities of 3 homeologs revealed by Sanger sequencing and 5B homeolog-specific SNP chip calls*





internal state



reward

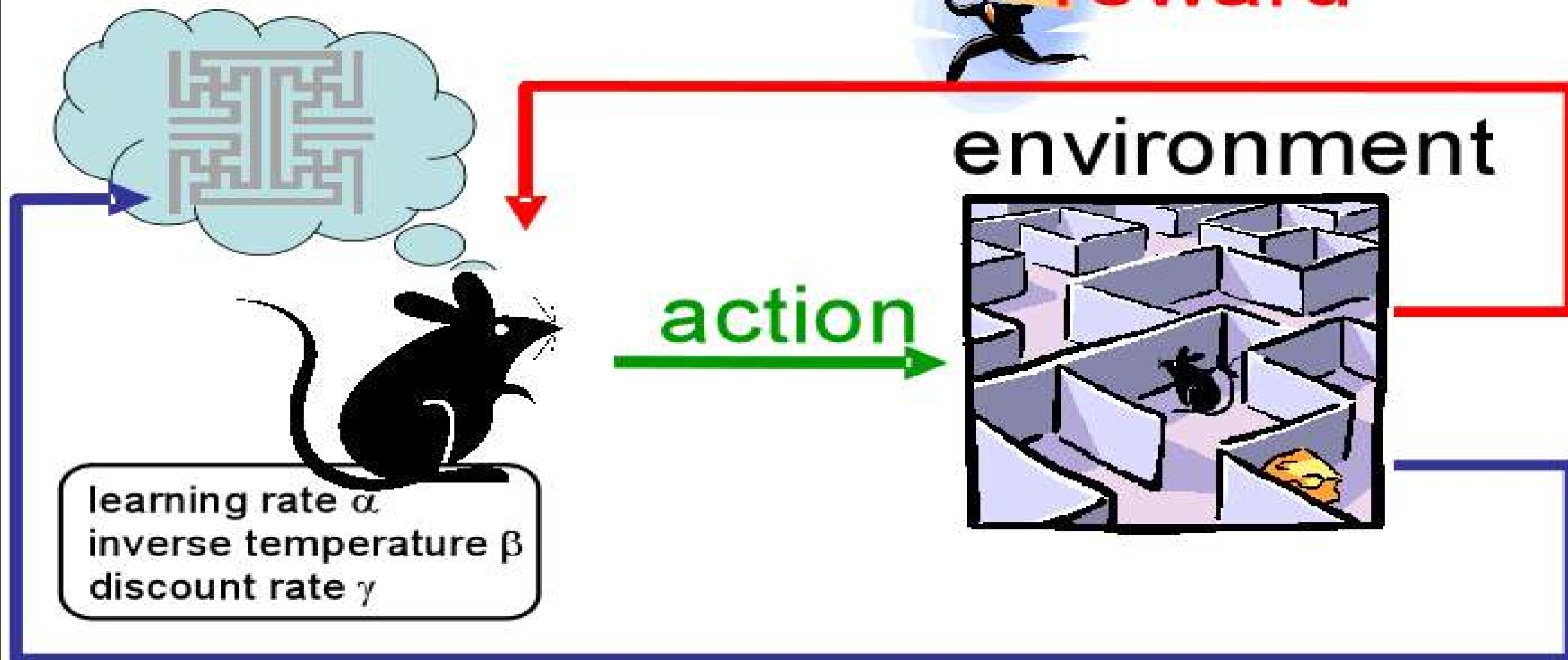
environment

action



learning rate α
inverse temperature β
discount rate γ

observation



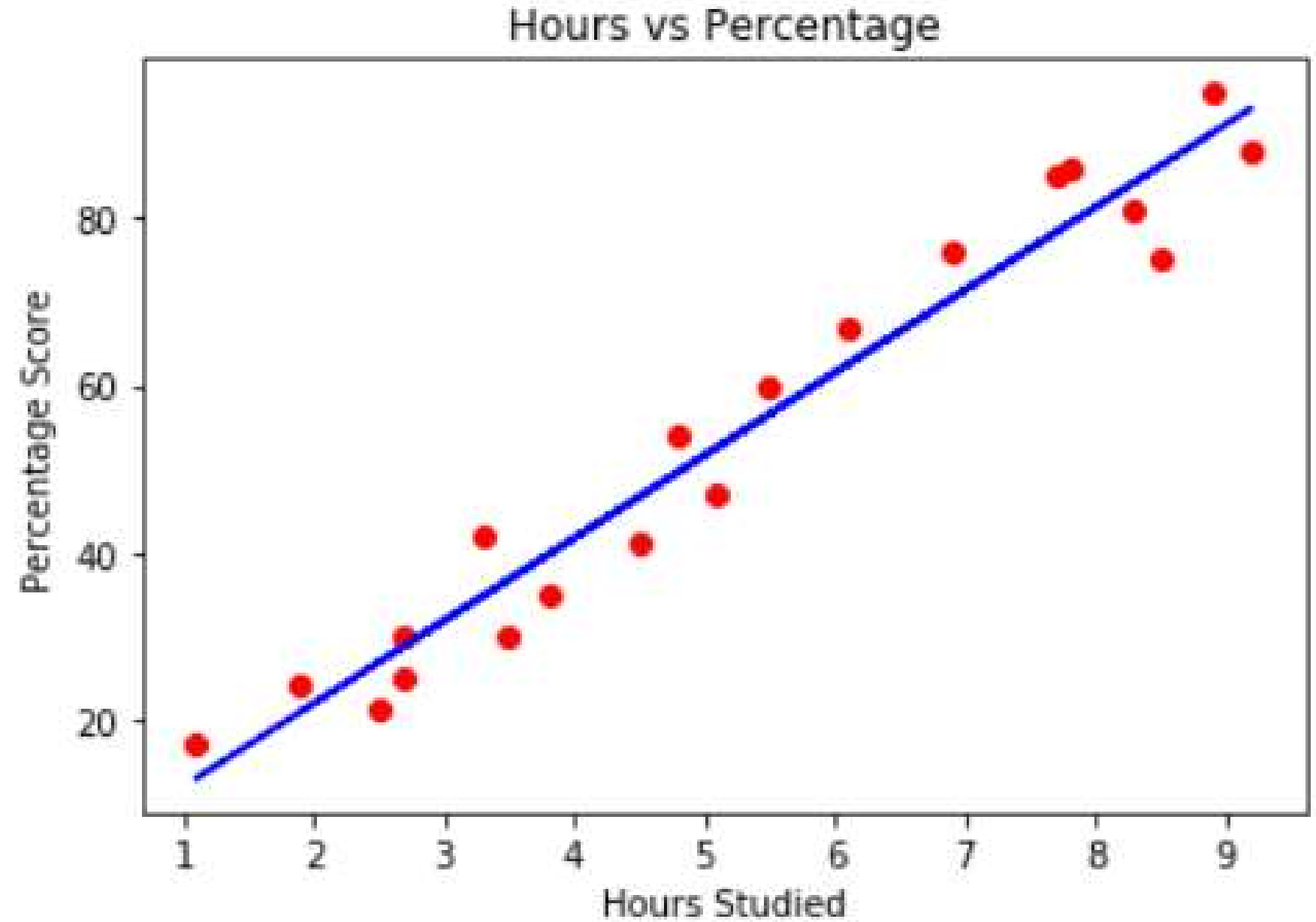
2016: World Go Champion Beaten by Deep Learning



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(tensorflow) C:\Users\fra_v>
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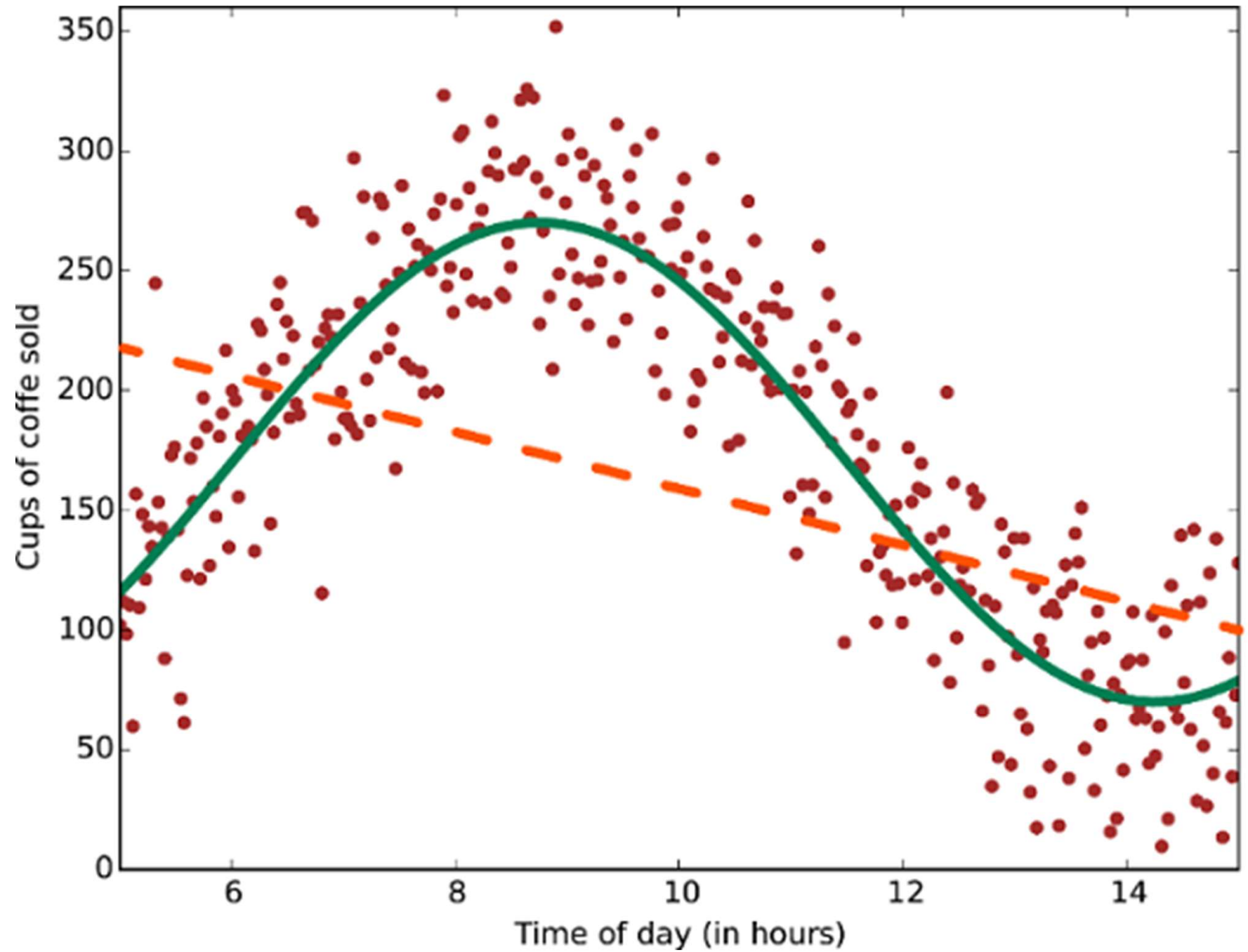
Machine Learning

- Curve fitting
(correlations) - linear



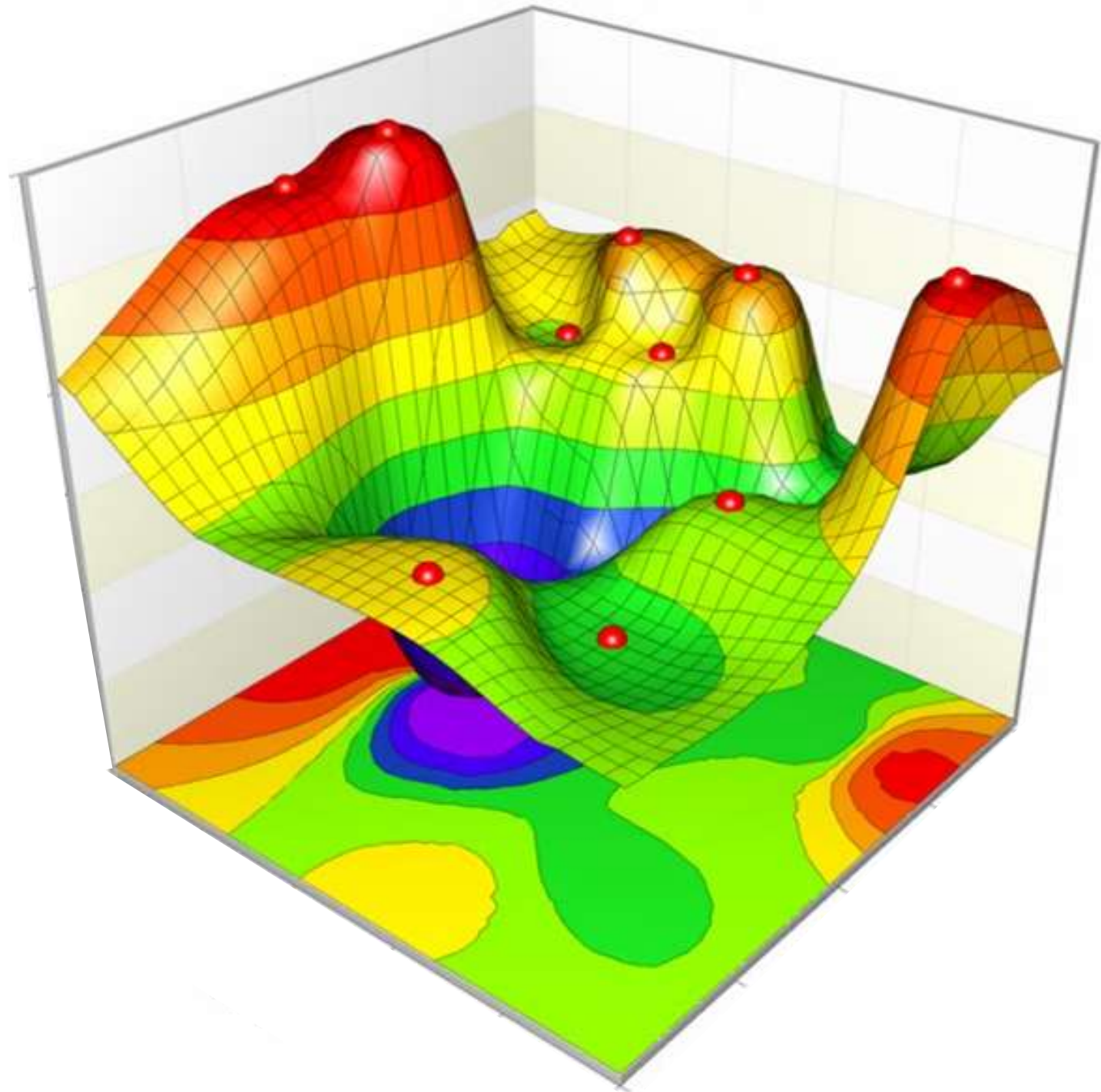
Machine Learning

- Curve fitting - nonlinear



Machine Learning

- Curve fitting - nonlinear



Machine Learning

- Deep Neural Networks

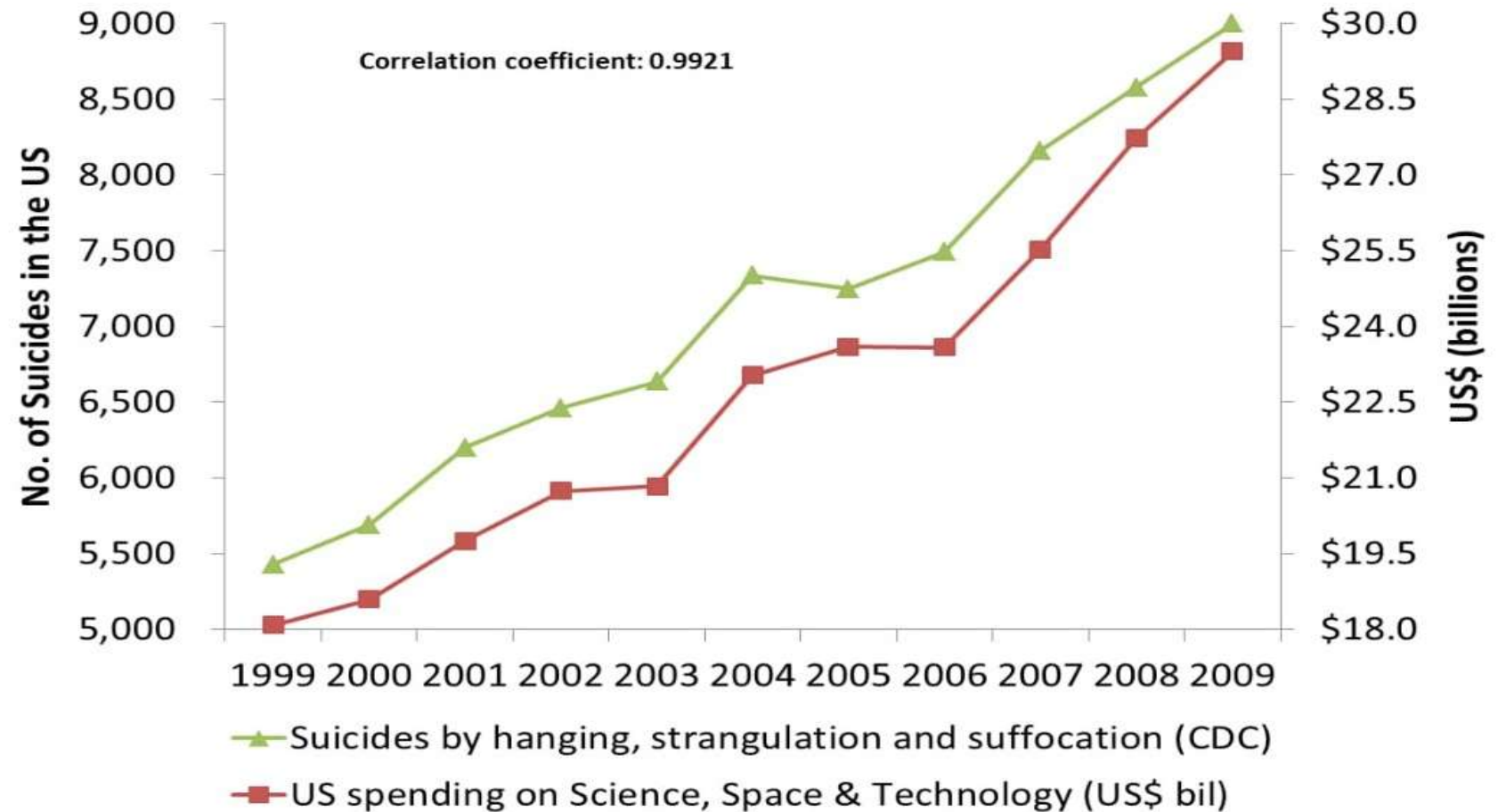


Highly dimensional, highly nonlinear
curve fitting

Machine Learning

- Spurious Correlations

Fitting can be highly misleading



Spurious Correlations



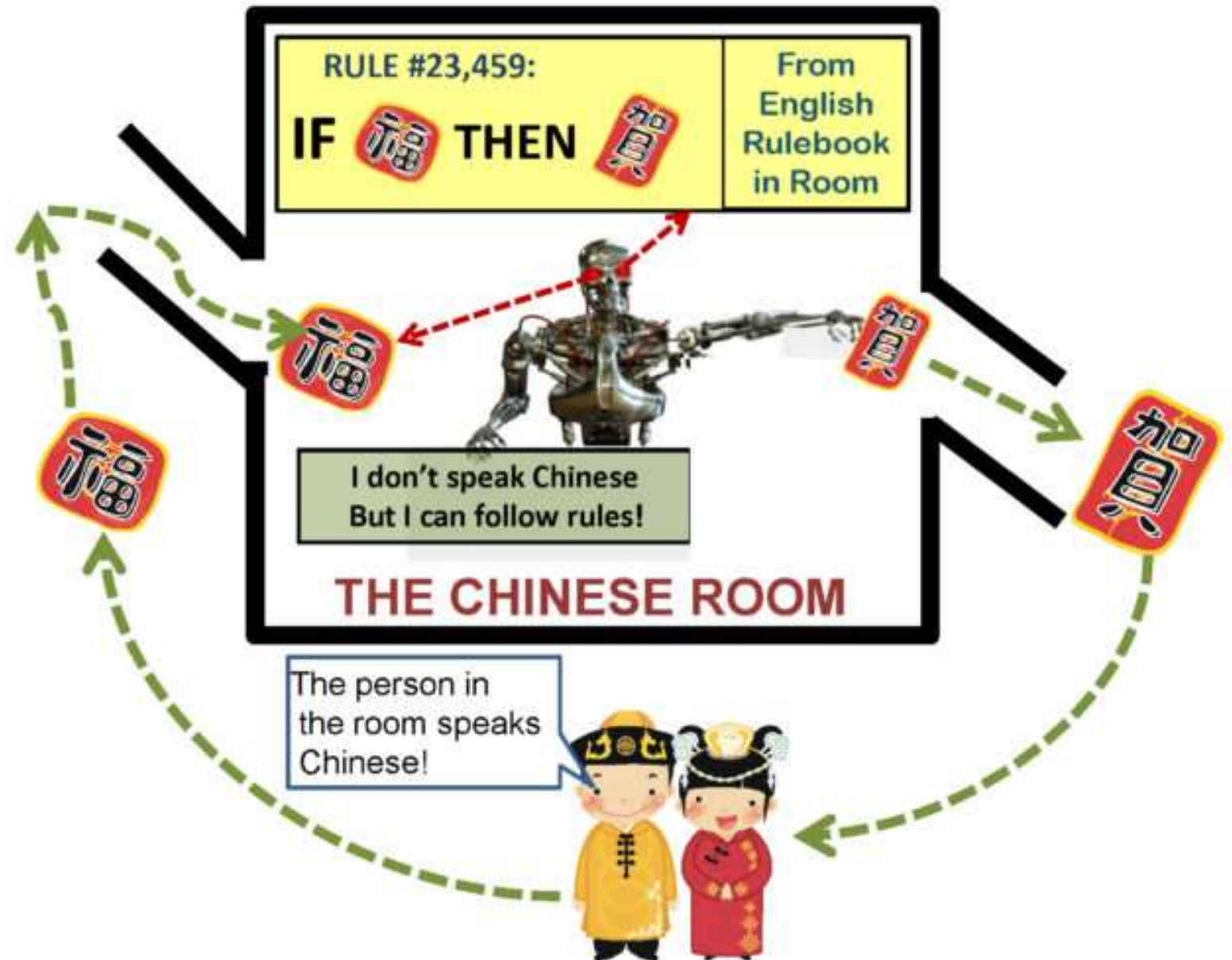
What just happened in artificial intelligence and how it is being misunderstood.

BY ADNAN DARWICHE

Human-Level Intelligence or Animal-Like Abilities?

"The vision systems of the eagle and the snake outperform everything that we can make in the laboratory, but snakes and eagles cannot build an eyeglass or a telescope or a microscope."

Fitting does not give us any understanding

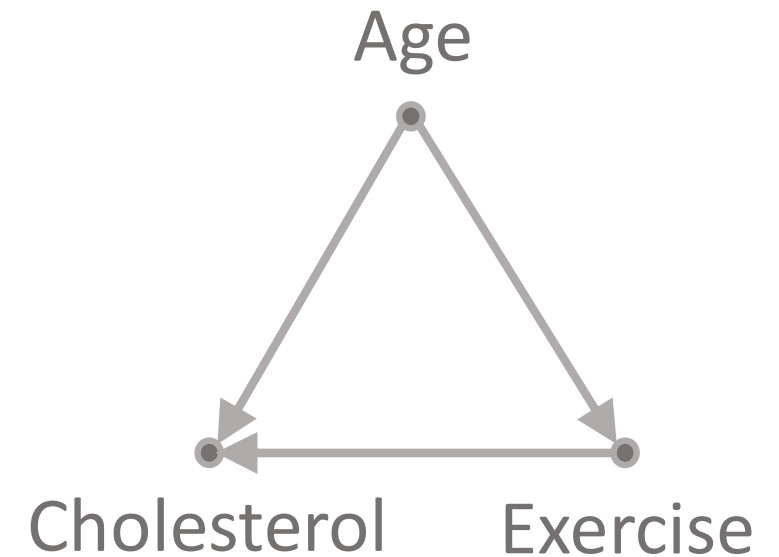
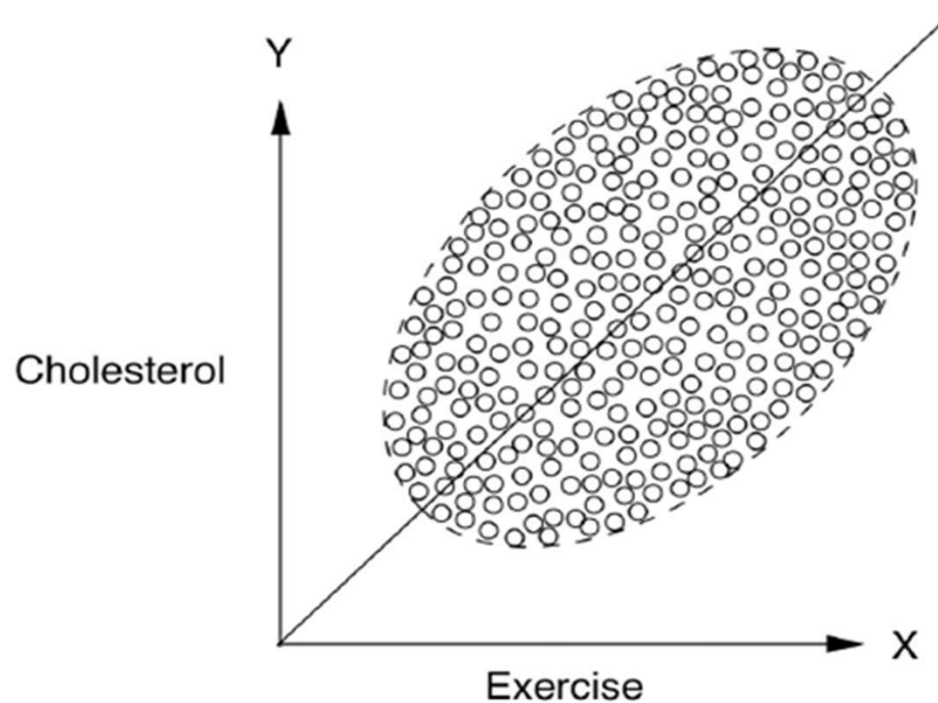
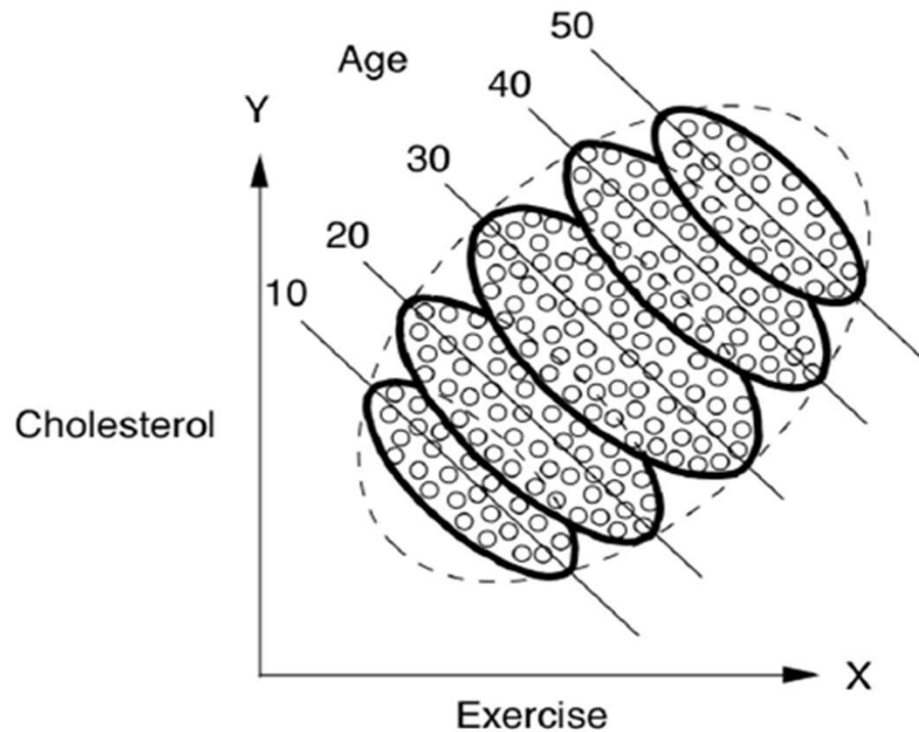


What can truly be achieved?

Does **Exercise** affects **Cholesterol**?

Simpson's Paradox

- No matter how much data you collect, the question can not be answered when using the data alone



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— *Judea Pearl*



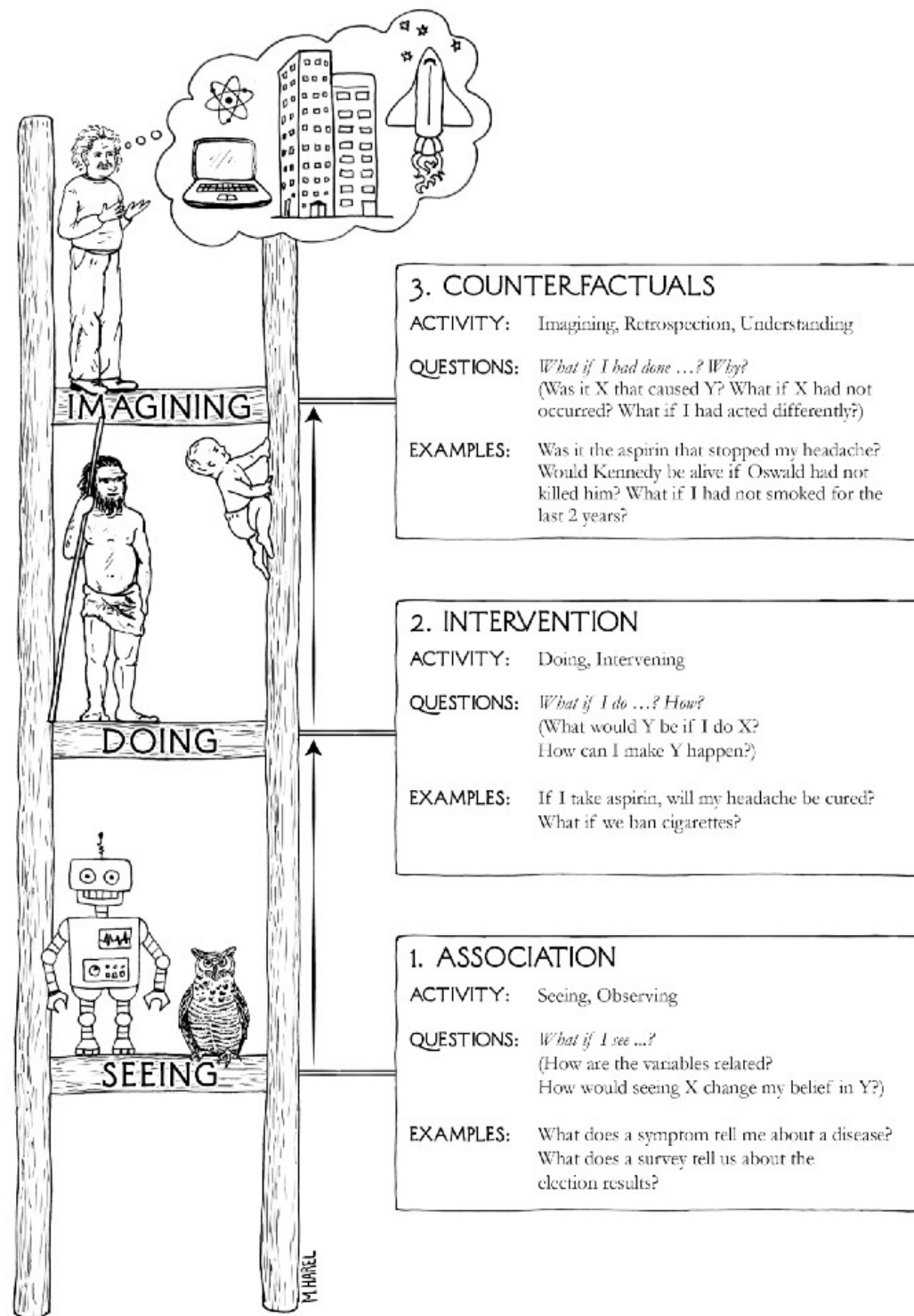


JUDEA PEARL
WINNER OF THE TURING AWARD
AND DANA MACKENZIE

THE BOOK OF WHY

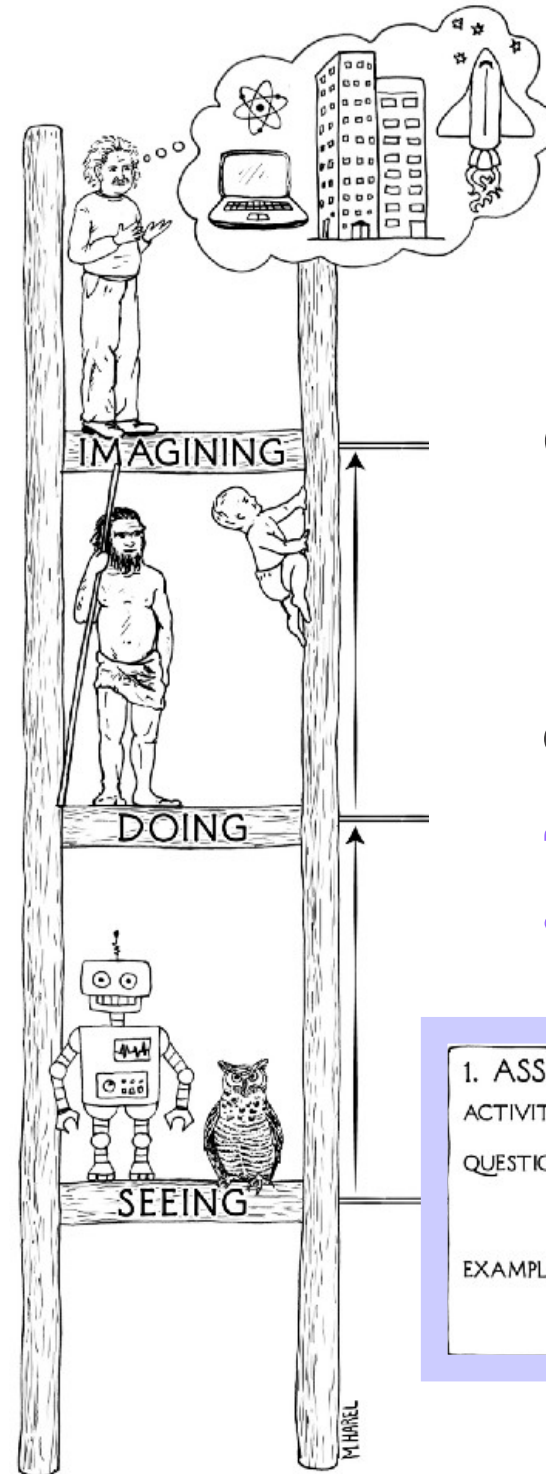


THE NEW SCIENCE
OF CAUSE AND EFFECT



The Ladder of Causation

Seeing; we are looking for regularities in observations.



“What if I see ...?”

Calls for predictions based on passive observations.

It is characterized by the question *“What if I see ...?”*

For instance, imagine a marketing director at a department store who asks,

“How likely is a customer who bought toothpaste to also buy dental floss?”

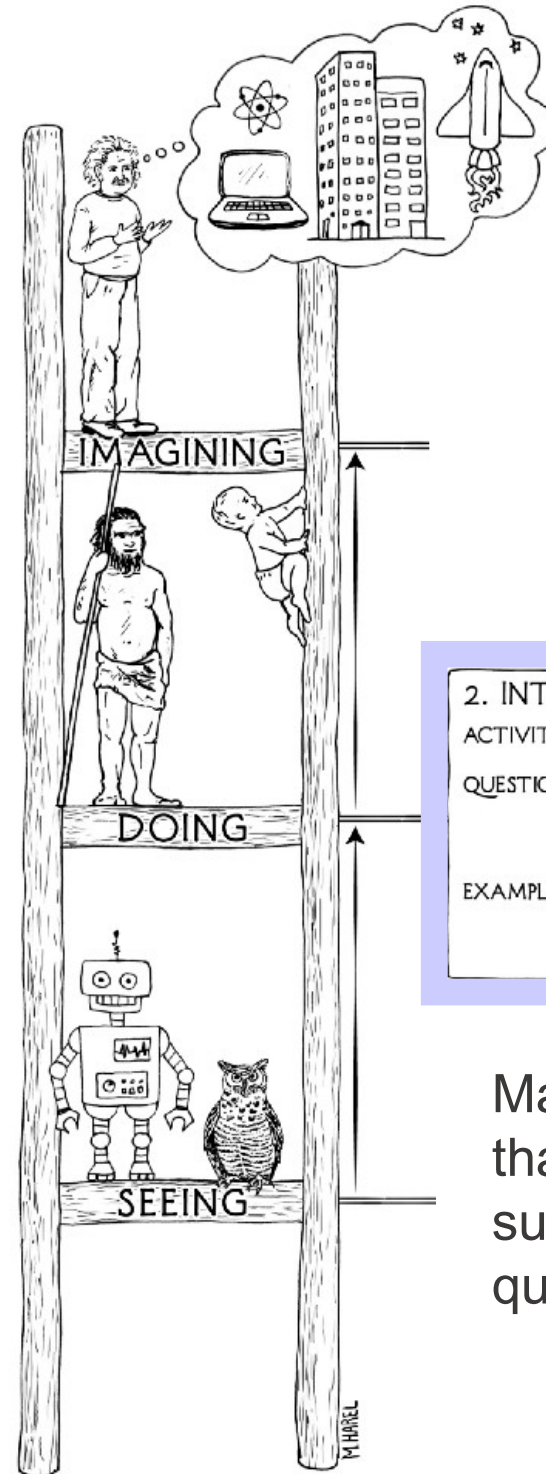
1. ASSOCIATION

ACTIVITY: Seeing, Observing

QUESTIONS: *What if I see ...?*
(How are the variables related?
How would seeing X change my belief in Y?)

EXAMPLES: What does a symptom tell me about a disease?
What does a survey tell us about the
election results?

Intervention; ranks higher than association because it involves not just seeing but changing what is.



“What if do ...?” & “How?”

We step up to the next level of causal queries when we begin to change the world. A typical question for this level is

“What will happen to our floss sales if we double the price of toothpaste?”

2. INTERVENTION

ACTIVITY: Doing, Intervening

QUESTIONS: *What if I do ...? How?*
(What would Y be if I do X?
How can I make Y happen?)

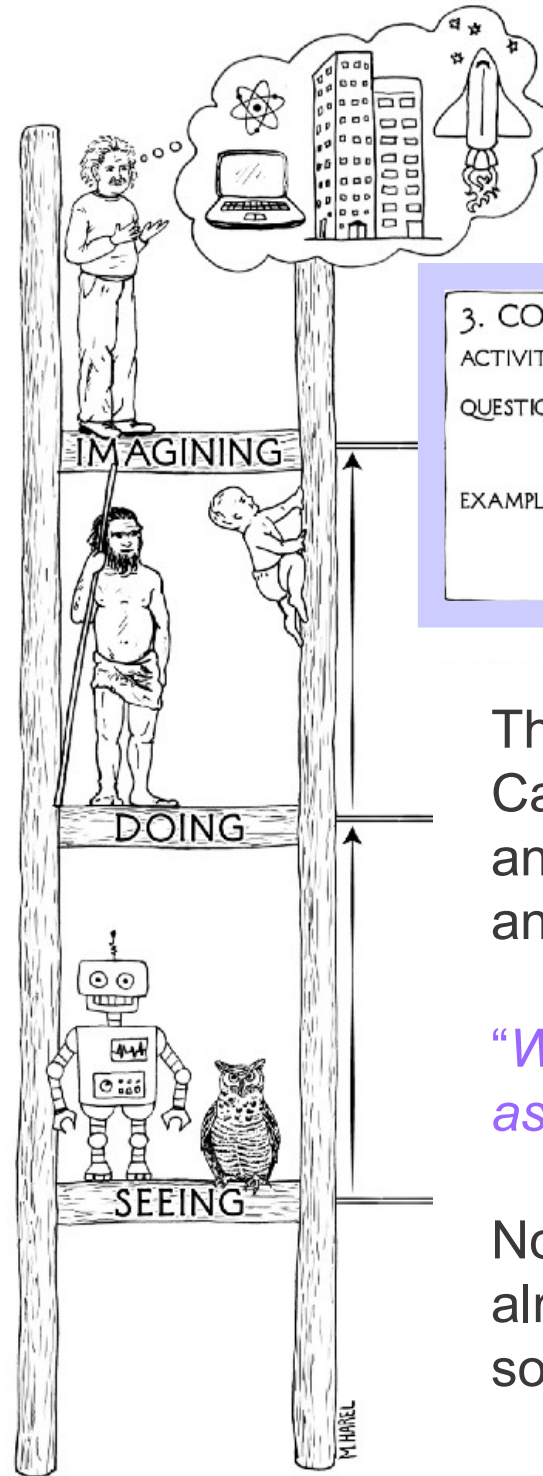
EXAMPLES: If I take aspirin, will my headache be cured?
What if we ban cigarettes?

This already calls for a new kind of knowledge, absent from the data, which we find at rung two of the Ladder of Causation, **Intervention**.

Many scientists have been quite traumatized to learn that none of the methods they learned in statistics is sufficient even to articulate, let alone answer, a simple question like

“What happens if we double the price?”

Counterfactuals; ranks higher than intervention because it involves **imagining, retrospection** and **understanding**.



“What if I had done ...?” & “Why?”

3. COUNTERFACTUALS

ACTIVITY: Imagining, Retrospection, Understanding

QUESTIONS: *What if I had done ...? Why?*
(Was it X that caused Y? What if X had not occurred? What if I had acted differently?)

EXAMPLES: Was it the aspirin that stopped my headache?
Would Kennedy be alive if Oswald had not killed him? What if I had not smoked for the last 2 years?

We might wonder, My headache is gone now, but

- *Why?*
- *Was it the aspirin I took?*
- *The food I ate?*
- *The good news I heard?*

These queries take us to the top rung of the Ladder of Causation, the level of **Counterfactuals**, because to answer them we must **go back in time, change history**, and ask,

“What would have happened if I had not taken the aspirin?”

No experiment in the world can deny treatment to an already treated person and compare the two outcomes, so we must import a whole new kind of knowledge.



CONTENTS

- **DATA TYPES**; to list different types of data and to learn hw they must be used for unsupervised learning.
- **DATA PREPROCESSING**; to preprocess data in such a way it can be used by unsupervised learning tasks,
- **CLUSTERING LEARNING**; to form homogeneous groups of observations and/or attributes using a given proximity measure,
- **CLUSTERING VALIDATION**; to evaluate and compare different clustering solutions to select the one to deploy.
- **ANOMALY DETECTION**; to find anomalous observations, to discover outliers observations, under different theoretical settings.
- **BAYESIAN NETWORKS**; to learn probabilistic/causal structure from data and to make decisions under uncertainty.